

Financial Mathematics Calculator

Mvelo Mnengi

June 13, 2026

Abstract

This report describes a program that computes standard actuarial annuity factors. Unlike printed actuarial tables, which only cover a fixed set of interest rates and integer numbers of periods, the program accepts any interest rate and any number of periods, including non-integer values. It covers level annuities and increasing annuities, in annuity-immediate, annuity-due, and continuous form.

1 Introduction

Actuarial and financial mathematics make heavy use of a small set of standard factors for valuing annuities: the present value and accumulated value of level and increasing payment streams, under the annuity-immediate, annuity-due, and continuous conventions. These are normally read from printed tables of a_n , $\ddot{a}_n$, s_n , $\ddot{s}_n$, and the corresponding increasing-annuity factors.

Printed tables, however, only cover a fixed grid of interest rates (e.g. 4%, 5%, 6%) and integer values of n , so any other rate or fractional period requires interpolation or manual recalculation. This project addresses that limitation by computing the requested factor directly for any interest rate i and any number of periods n supplied by the user.

2 Objectives

1. Provide separate categories for level annuity factors and increasing annuity factors.
2. For each category, provide the present value and accumulated value of the annuity-immediate, annuity-due, and continuous versions.
3. Accept an arbitrary interest rate i and number of periods n , including non-integer n .
4. Validate all user input.

3 Methodology

The program presents a main menu with three options: level annuities, increasing annuities, and exit. Selecting either of the first two opens a sub-menu listing the specific factors in that category. Once a factor is selected, the program requests the interest rate i and number of periods n , computes the factor, and returns the result.

Input is validated at each stage, so that invalid menu selections or numerical entries do not cause the program to terminate unexpectedly.

4 Formulas

Throughout, i is the interest rate per period, n is the number of periods, and $\delta = \ln(1 + i)$ is the equivalent force of interest.

4.1 Level Annuities

For an annuity-immediate paying 1 at the end of each period, the present value and accumulated value are

$$a_n = \frac{1 - (1 + i)^{-n}}{i}, \quad s_n = \frac{(1 + i)^n - 1}{i}. \quad (1)$$

The annuity-due versions, with payments at the start of each period, follow from the usual due adjustment:

$$\ddot{a}_n = a_n(1 + i), \quad \ddot{s}_n = s_n(1 + i). \quad (2)$$

For a continuous annuity paying at rate 1 per unit time,

$$\bar{a}_n = a_n \cdot \frac{i}{\delta}, \quad \bar{s}_n = s_n \cdot \frac{i}{\delta}. \quad (3)$$

4.2 Increasing Annuities

For an annuity-immediate with payments $1, 2, 3, \dots, n$, the present value and accumulated value are

$$(Ia)_n = \frac{a_n - n(1 + i)^{-n}}{i}, \quad (Is)_n = \frac{s_n - n}{i}. \quad (4)$$

The annuity-due versions are

$$(I\ddot{a})_n = (Ia)_n(1 + i), \quad (I\ddot{s})_n = (Is)_n(1 + i). \quad (5)$$

For the continuously increasing continuous annuity,

$$(\bar{I}\bar{a})_n = \frac{\bar{a}_n - n(1 + i)^{-n}}{\delta}, \quad (\bar{I}\bar{s})_n = (\bar{I}\bar{a})_n(1 + i)^n. \quad (6)$$

5 Testing and Verification

Each factor was computed for $i = 0.05$, $n = 10$ and checked against standard actuarial tables and the algebraic relationships between factors (e.g. $\ddot{a}_n = a_n(1 + i)$ and $\bar{a}_n = a_n \cdot i/\delta$).

Factor	Value
a_{10}	7.721735
s_{10}	12.577893
$\bar{a}_{10}$	7.913209
$\bar{s}_{10}$	12.889783
$(Ia)_{10}$	31.652048
$(\bar{I}\bar{a})_{10}$	36.361346
$(\bar{I}\bar{s})_{10}$	59.228802

Table 1: Factor values at $i = 0.05$, $n = 10$.

To check the handling of non-integer periods, a_n was also computed for $i = 0.05$, $n = 10.5$, giving $a_{10.5} = 8.017640$, a value that a printed table (which only lists integer n) could not provide directly.

6 Conclusion

The calculator computes level and increasing annuity factors directly from their closed-form formulas, for any interest rate and any number of periods, including non-integer values, without needing interpolation. It provides the annuity-immediate, annuity-due, and continuous versions of each factor.

Possible extensions include deferred annuities, perpetuities, and varying annuities with geometric rather than arithmetic growth.